Problem Statement

Flood disaster response demands rapid, reliable intelligence — but critical operational signals remain split across disconnected tools.

01

Fragmented Remote Sensing Tools

- SAR flood mapping, building damage, and task planning operate in isolated tools.
- No unified pipeline connects pixels to rescue.

02

Latent & Cloud-Impaired Mapping

- Optical sensors fail under storm clouds/rain.
- Manual photo-interpretation causes delays during the golden 72-hour window.

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THE OPERATIONAL RESPONSE GAP

Detection ≠ Assessment ≠ Explanation ≠ Action

A first responder requires one continuous, explainable workflow connecting multi-sensor satellite pixels directly to prioritized, resource-optimized rescue missions.

03

Opaque Black-Box AI Predictions

- Deep models lack visual attribution (Grad-CAM), causing operator skepticism in AI decisions.
- Missing audit trails prevent defensible triage.

04

Non-Actionable Hazard Outputs

- Raw hazard maps fail to specify rescue tasks.
- Ignores finite squad counts, equipment limits, and transit constraints across flooded zones.

INTEGRATED FRAMEWORK

1 **Flood Extent Mask** — Where is water? (SAR U-Net)

→ 2 **Damage Severity** — What is affected? (xBD Ensembles)

→ 3 **Explainable XAI Audit** — Why this prediction? (Grad-CAM)

→ 4 **Optimized Rescue Dispatch** — What needs action first? (18-Team)